

ATHABASCA AGCM, AB, Canada

WMO: 712710

Lat: 54.6347N

Lon: 113.3819W

Elev: 2083

StdP: 13.62

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -23.7	(c) -17.6	(d) -30.0	(e) 1.1	(f) -23.2	(g) -24.0	(h) 1.6	(i) -17.7	(j) 17.9	(k) 32.8	(l) 15.8	(m) 31.6	(n) 2.4	(o) 220	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	81.2	63.0	78.4	61.6	75.6	60.5	66.0	77.2	64.1	74.7	62.4	72.5	5.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.7	88.8	71.7	59.8	83.0	69.2	58.1	78.0	67.2	31.9	77.4	30.4	74.8	29.1	72.7	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	13.9	12.2	-31.1	86.9	6.0	1.9	-35.4	88.3	-38.9	89.4	-42.3	90.5	-46.7	91.9
			-31.3	69.9	6.0	2.6	-35.6	71.8	-39.1	73.3	-42.4	74.8	-46.8	76.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	13.5	13.4	24.3	38.5	51.7	58.4	62.4	60.6	51.9	39.3	23.7	10.5
	DBStd	21.76	16.39	14.88	13.93	9.16	8.47	5.70	4.70	5.77	8.34	8.56	12.28	14.79
	HDD50	5838	1130	1025	798	358	83	8	0	2	76	344	790	1224
	HDD65	10113	1595	1445	1263	796	418	208	108	158	397	796	1240	1689
	CDD50	1266	0	0	0	12	135	259	385	331	132	12	0	0
	CDD65	66	0	0	0	0	4	10	28	22	2	0	0	0
	CDH74	940	0	0	0	4	102	148	335	253	96	2	0	0
	CDH80	130	0	0	0	0	16	13	48	37	16	0	0	0
Wind	WSAvg	5.9	5.8	6.0	5.8	6.6	6.5	5.9	5.1	5.0	5.3	6.8	5.8	6.0
Precipitation	PrecAvg	17.0	0.9	0.4	0.7	1.0	1.9	3.2	3.2	2.2	1.4	0.7	0.8	0.7
	PrecMax	21.2	3.7	0.9	1.5	1.9	3.1	5.3	6.3	4.6	3.1	1.3	2.8	1.3
	PrecMin	12.6	0.0	0.0	0.1	0.0	0.3	1.3	1.5	0.9	0.2	0.1	0.1	0.0
	PrecStd	2.8	0.9	0.3	0.4	0.5	0.7	1.1	1.2	0.9	0.7	0.4	0.7	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.5	47.4	55.2	72.2	81.9	81.7	84.8	83.9	82.6	70.1	53.5	43.9
		MCWB	38.5	39.0	43.4	51.5	57.0	62.1	66.6	64.7	62.3	53.9	44.3	36.1
	2%	DB	43.3	41.7	48.9	64.2	76.7	78.1	80.7	80.1	76.4	61.6	46.5	39.2
		MCWB	36.6	35.1	39.6	46.8	55.2	60.0	64.6	64.3	60.1	48.8	40.1	33.4
	5%	DB	39.9	37.9	44.9	58.3	71.9	75.0	77.8	76.5	71.7	56.9	42.1	34.6
		MCWB	34.5	33.2	36.9	43.8	52.9	58.5	63.0	62.6	57.8	46.0	36.8	30.5
	10%	DB	36.2	33.8	41.3	53.4	68.2	71.6	74.7	73.5	66.6	52.3	38.1	30.7
		MCWB	32.1	30.1	34.6	41.6	50.9	57.8	61.6	61.1	54.8	43.4	34.1	27.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	39.7	43.4	51.8	59.1	65.4	69.6	68.3	63.7	54.7	45.1	36.4
		MCDB	45.4	46.8	54.0	71.6	75.5	76.6	80.4	79.6	78.9	68.9	52.3	42.5
	2%	WB	37.0	35.9	40.2	47.5	56.8	62.6	66.5	65.8	61.0	50.4	40.7	33.5
		MCDB	42.4	40.6	47.8	61.9	72.6	74.2	77.3	77.3	73.4	59.6	45.6	38.5
	5%	WB	34.7	33.2	37.4	44.7	54.6	60.5	64.5	63.8	58.2	46.8	37.2	30.5
		MCDB	39.5	37.6	44.2	56.4	69.3	71.6	74.9	74.2	69.8	55.3	41.5	34.6
	10%	WB	32.1	30.5	34.9	42.2	52.2	58.5	62.8	62.0	55.8	44.3	34.4	27.4
		MCDB	36.3	34.3	40.7	52.0	65.8	68.8	72.4	71.5	65.7	50.7	37.7	30.3
Mean Daily Temperature Range	5% DB	MDBR	15.6	17.2	18.0	18.6	22.6	20.5	20.6	20.9	20.9	16.2	13.1	14.3
		MCDBR	16.0	18.8	20.2	27.0	28.6	25.7	25.8	25.6	29.6	24.0	17.0	17.6
		MCWBR	11.6	14.1	12.4	14.2	13.1	11.7	13.0	12.9	15.4	13.8	11.7	14.1
	5% WB	MCDBR	15.5	18.0	19.4	24.8	26.5	23.3	23.4	23.4	28.4	22.4	16.0	17.5
		MCWBR	11.7	14.0	12.2	13.7	12.9	11.7	12.7	12.7	15.3	13.5	11.7	14.1
Clear-Sky Solar Irradiance	taub	0.251	0.272	0.300	0.324	0.338	0.351	0.353	0.360	0.306	0.281	0.250	0.231	
	taud	2.289	2.304	2.334	2.400	2.418	2.386	2.435	2.360	2.530	2.568	2.461	2.302	
	Ebn at Noon	227	262	279	287	288	284	281	272	277	261	233	212	
	Edh at Noon	17	24	30	32	34	36	33	34	25	19	15	14	
All-Sky Solar Radiation	RadAvg	259	579	1055	1470	1792	1867	1881	1543	1065	592	273	186	
	RadStd	29	42	82	117	155	127	102	110	109	82	30	15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page